

FRC Scouting App — Project Handoff Summary (v10)

A concise, self-contained handoff for the offline FRC scouting desktop app. Covers the project goal, architecture, what's been built and verified through Stage 16, and exactly what a future session needs to pick this up with no access to prior chat history. **This document supersedes the v9 PDF** — it is current through Stage 16 (Receipt Content Trim + Cut-off Fix). If this document and the code ever disagree, the code is ground truth.

1. Project Goal

A fully offline Python desktop app (Windows + Mac) that replicates a complex FRC scouting Google Sheet: SQLite local storage only, zero network calls, four required UI tabs in order (Match Lookup, Team Lookup, Pick List, Rankings) matching the sheet's first four tabs, and a custom non-Monte-Carlo skew-distribution statistical method for forecasting robot performance and match outcomes. Correctness has been explicitly prioritized over speed throughout. Two additions beyond the original four-tab requirement now exist: a fifth tab, **Match Simulator** (Stage 14), and a **Print Results** button on Match Lookup (Stage 15, refined in Stage 16).

2. Source Material

- **The Google Sheet** — “Copy of Worlds Scouting Sheet”, 32 tabs, ~215,687 cells. First four tabs match the four required UI tabs; the other 28 are backend calculation sheets.
- **Formula export** — `Copy_of_WorlDs_Scouting_Sheet_-_FORMULA_EXPORT.csv`, produced by a resumable Apps Script, is the ground-truth source for every formula ported to Python and for every bug fix described below. **Required for any future calculation work** — keep it.
- **The algorithm PDF** — “Scouting Algorithm Overview” describes the skew-distribution method: pick a symmetrizing transform per team, fit a normal PDF on the transformed axis, rescale peak to 1 with a percentile cutoff, then combine three teammates' curves and compare two alliances to estimate win probability.
- **Missing “Super Scouting” sheet** — referenced by a few Team Lookup fields (team name, weak-to-defense notes, climb-confidence flags) but does not exist in the export. Confirmed by the user as not needed; those fields are intentionally left unimplemented.

3. Current File Structure

Delivered as `frc_scouting_app/`, most recently zipped as **frc_scouting_app_18.zip** (Stage 16). Every `.py` file's docstring is the authoritative spec for what it implements.

Path	Role / last touched
<code>main.py</code>	Entry point. Unaffected since original build.
<code>database.py</code>	SQLite schema. Unaffected since original build.
<code>data_import.py</code>	CSV importers (<code>manual_raw_data</code> , <code>match_schedule</code>). Unaffected since Stage 9.
<code>requirements.txt</code>	matplotlib only, sole pip dependency (Stage 13). Stage 15 added an explanatory comment, no new dependency.
<code>README.md</code>	Setup, data-import format reference, full stage-by-stage log — kept in sync every stage, now through Stage 16.
<code>calculations/</code>	All backend math (32 sheets ported). Untouched by Stages 14, 15, and 16, in every file, no exceptions.
<code>ui/app.py</code>	Main window / tab notebook / Data menu. Stage 14 added the Match Simulator tab; Stage 15 added “Select Receipt Printer...”. Unaffected by Stage 16.

Path	Role / last touched
ui/match_lookup_tab.py	Tab 1: Match Lookup. Stage 14 moved panel/chart code out to alliance_display.py. Stage 15 added the "Print Results" button + <code>_print_results()</code> . Unaffected by Stage 16 (Stage 16 only touched printing/escpos_builder.py).
ui/alliance_display.py	Shared panel/chart code (Stage 14) used by Match Lookup and Match Simulator.
ui/match_simulator_tab.py	Tab 5: Match Simulator (Stage 14). No Print Results button (out of scope for Stage 15/16 — see Section 8).
ui/team_lookup_tab.py, pick_list_tab.py, rankings_tab.py	Tabs 2–4. Unaffected since Stage 11 (Team Lookup) / original build (the other two).
printing/	Stage 15 package (printer discovery + ESC/POS byte building), refined in Stage 16. See Section 7.
data/scouting.db, data/pr inter_settings.json	Runtime SQLite file and remembered-printer JSON (created automatically; the latter deliberately lives outside scouting.db so it survives a "New competition" import).

4. Runtime / Environment

- Python 3.11+, tkinter (ships with the official python.org installer — on Mac must be that installer, not a Tk-less Homebrew build).
- Local storage: SQLite via the stdlib sqlite3 module, one file at data/scouting.db.
- Sole pip dependency: matplotlib (lazily imported inside chart-drawing functions since Stage 13). numpy/scipy/pandas were removed in Stage 13 and never reintroduced.
- Printing (Stage 15) needs no pip package on either OS: Windows uses ctypes + winspool.drv; macOS shells out to lp/lpstat (CUPS). Only the printer's own OS-level driver needs installing.
- Zero network calls anywhere in the codebase, by design.
- Setup: `pip install -r requirements.txt` then `python main.py`.

5. Stage History (Summary)

Full narrative detail for Stages 1–15 lives in the v9 PDF and in README.md's stage-by-stage log; it is condensed here to keep this handoff document focused. Nothing in Stages 1–15 changed as a result of Stage 16.

Stage	What it delivered
1–3	Full calculation pipeline ported: raw data import, skew-curve fitting, defense/compatibility stats, power rankings, alliance combination, match odds, Pick List, Rankings.
4	Two-CSV import model built around the user's real data format; fixed a zero-collapse bug in Adjusted Score.
5	Three real bugs found/fixed between Score Variance and Adjusted Score Variance, verified cell-by-cell against the formula export.
6	Experiment (user-requested): Adjusted Score Variance removed from the pipeline; plain Score Variance used everywhere instead. Still active in production.
7	Two real bugs fixed in alliance_generator.py (branch ordering; alliance-level axis truncated too low for 3-robot scores). Mean absolute error dropped ~175→~100 pts; win-direction accuracy 67%→69%.
8	Match Lookup / Team Lookup dropdowns replaced with digit-validated Entry fields; Rankings got a Sort-by dropdown.
9	Fixed new-competition imports silently mixing with old data; added an explicit Update-current vs. New-competition import choice.
10–11	Team Lookup: added a Score Trend chart, then removed 3 redundant panels so it fits on screen.
12	Match Lookup: added a smoothed Score Distribution chart overlaying both alliances' curves.

Stage	What it delivered
13	Startup performance: replaced a hot <code>scipy.stats.norm.pdf()</code> call with a small <code>stdlib</code> function; dropped <code>numpy/scipy/pandas</code> ; made <code>matplotlib</code> import lazy. <code>build_all()</code> dropped $\sim 0.63s \rightarrow \sim 0.02s$, output unchanged to floating-point noise.
14	New fifth tab, Match Simulator — type any six team numbers, see the same odds/score output as a real match minus the actual score. Panel/chart code factored into a shared <code>ui/alliance_display.py</code> so Match Lookup and Match Simulator can't visually drift apart.
15	"Print Results" button on Match Lookup: two 80mm thermal receipts (Red, then Blue) via a new printing/ package, no new pip dependency on either OS. Full detail in Section 7.
16	Latest. Trimmed the receipt content to be shorter/more direct, and fixed a cut-off bug. <code>printing/escpos_builder.py</code> only. Full detail in Section 8.

Known open item carried since Stage 6: plain Score Variance (not Adjusted) is live in production — see Section 9's next steps to revisit if Pick List quality looks off. Known open item since Stage 8: a qualitative Match-Odds comparison against the live Google Sheet was "passable but not identical"; no specific numbers were logged. Known open item since Stage 13: all 5 tabs still eagerly `refresh()` at startup before the user sees any but the first.

6. Current UI State (all 5 tabs fully wired, no placeholders)

- **Match Lookup** — type/enter a match number; shows both alliances (Score Variance avg/RRP, actual score, live expected score + win probability) and the Score Distribution chart. "Print Results" button next to "Go" (Stage 15/16 — Section 7).
- **Team Lookup** — Score Trend chart plus 4 stat panels. Unaffected since Stage 11.
- **Pick List / Rankings** — unaffected since original build / Stage 8 (Sort-by dropdown).
- **Match Simulator** — six digit-only team-number fields (3 red, 3 blue) + Simulate button. Duplicate-team entries are rejected. Shows the same panels/chart as Match Lookup, minus the actual-score line. No Print Results button.
- **Data menu** — Import Match Schedule / Import Raw Data / Set My Team Number / Recalculate Now / **Select Receipt Printer...** (Stage 15).

7. Stage 15 Recap — Match Results Printing

A "Print Results" button on Match Lookup sends two 80mm ESC/POS thermal receipts (Red, then Blue) to a USB receipt printer. Key design points, unchanged by Stage 16:

- **Printer discovery goes through the OS print spooler** (not raw USB/vendor IDs) — foolproof with multiple USB devices plugged in, since the OS already knows the printer by name once its driver is installed.
- **Auto-resolve logic** (`printing/print_dialog.py::_resolve_printer()`): use the remembered printer if still installed → auto-pick-and-remember if exactly one printer is installed and none is remembered → show a picker dialog if ambiguous → plain-language error if zero printers are installed. Reselectable anytime via Data → Select Receipt Printer....
- **Cross-platform, zero new pip dependencies**: Windows via `ctypes` + `winspool.drv`; macOS via `subprocess` + `lp/lpstat` (CUPS).
- **No calculation logic in printing/** — every number comes from the same `ComputedData` / `MatchOddsResult` already computed for the on-screen panels.
- **Not yet confirmed**: an actual receipt printed by real hardware, and the picker dialog in a real Tkinter window — the build environment used has no display and no printer. Top priority for the next session with real hardware (Section 10).

8. Stage 16 — Receipt Content Trim + Cut-off Fix (Latest Stage)

Request: shorten and simplify the printed receipt content. Team Lookup's on-screen panels must not change — this is a printout-only change. **Scope: printing/escpos_builder.py only.** No file under calculations/ or ui/ was touched.

Fields removed from each team's printed section:

- **Auto Balls Scored** — removed RRP and Matches. Kept: Average, Std Dev.
- **Teleop Balls Scored** — removed RRP and Matches. Kept: Average, Std Dev.
- **Average Compatability** — entire section removed from the receipt.
- **Overall Power** — removed ROP, RDP, and Distance from (1,1). Kept: RPP, Weak-to-defense flag.

Unchanged on the receipt: Score history, Fitted Score Distribution (mode + 95th-percentile cutoff), and the alliance's match prediction (actual score / expected score / win probability).

Cut-off fix:

The Win probability line was getting clipped by the paper cut on some printers. Fixed by adding a few extra blank feed lines immediately before the cut command, on top of the trailing feed cut () already sent.

Validation performed this session:

- `ast.parse()` syntax check on the edited file.
- `build_alliance_receipt()` re-exercised against synthetic team data; asserted none of the six removed fields (RRP/Matches under Auto and Teleop, the whole Average Compatability section, ROP/RDP/Distance) appear anywhere in the output bytes.
- Visual inspection of the decoded receipt text confirmed the trimmed sections read cleanly and several blank lines now precede the cut command.

Not yet confirmed (same standing limitation as Stage 15): whether the added blank-line margin is actually enough on the user's specific printer model — this can only be confirmed by printing a real receipt. If it's still clipped, or now leaves too much blank paper, the fix to adjust is the small `rb.feed(4)` call right before `rb.cut( )` at the end of `build_alliance_receipt()` in `printing/escpos_builder.py`.

Delivered as **frc_scouting_app_18.zip**. README.md's "Stage 16" section was added in the same session to stay in sync with this PDF.

9. Testing Performed (Cumulative Summary)

- **Stages 1–13:** headless synthetic-data smoke tests after every stage; cell-by-cell cross-checks against the formula-export CSV (Stage 5); 110-match mean-absolute-error and win-direction comparisons (Stage 7); real-machine confirmation of every UI/chart change through Stage 13 ("latest results look good").
- **Stage 14:** `ast.parse` check, line-by-line refactor review, a fresh headless pipeline run on a 45-team/90-match competition's real CSVs, and a direct headless call to the alliance/odds functions on an arbitrary six-team combination.
- **Stage 15:** `ast.parse` across the project; unit tests of the word-wrap and label/value-row helpers; `build_alliance_receipt()` exercised against a fully-populated team, a no-data team, and the odds-unavailable case; `printer_backend.py` confirmed to fail cleanly (not crash) on an unsupported OS; `settings.py` persistence round-tripped on disk; the four-branch printer-resolution logic tested with mocks; `_print_results()` exercised end-to-end against a real temporary SQLite database with the printer backend mocked.
- **Stage 16:** see Section 8.
- **Standing gap across Stages 8–16:** the build environment used for every session has had no display and (for Stage 15/16) no USB printer, so the Tkinter GUI and physical receipts have never been seen by human eyes in this project's own testing — only by the user, on their own machine, at their own discretion between sessions.

10. Suggested Next Steps (Priority Order)

- **Top priority:** real-hardware confirmation of Stage 15/16 printing — plug in a real USB 80mm thermal printer on a real Windows or Mac machine and confirm the button, the auto-resolve/picker flow, both receipts printing legibly with the win-probability line no longer clipped, and Data → Select Receipt Printer... working.
- Real-Tkinter-window confirmation of Stage 14 (five tabs in order, Match Simulator's fields/validation, Match Lookup unchanged after its refactor) — still outstanding since Stage 14.
- Use a second competition's data (ideally with a live Google Sheet to compare against) to make progress on the bookmarked Match-Odds validation from Stage 8/13 — this time logging specific match numbers and both systems' percentages.
- Keep an eye on real-world Pick List / Rankings quality now that plain Score Variance (Stage 6) is in production use; the Adjusted-Score-Variance revert path documented in the v9 PDF's Section 10 is still available if needed.
- Consider deferring non-visible tabs' refresh() until first shown (open since Stage 13) — needs a real display to verify safely.
- Optional polish, not yet requested: a Print Results button on Match Simulator (would reuse printing/ as-is, omitting actual_score); a predictions-vs-actual-outcomes history view; a visible “current competition” label; a matplotlib NavigationToolbar2Tk on either chart; an on-screen “test print” option.

11. How to Resume This Project in a Fresh Session

Provide:

- This PDF (v10).
- Copy_of_Worlds_Scouting_Sheet_-_FORMULA_EXPORT.csv — required for any session touching calculations/; not required to re-verify Stages 13–16.
- The current project folder / zip — **frc_scouting_app_18.zip** (or a newer-numbered zip if one has been shared since).
- The user's real Raw Data / Match Schedule CSVs for whichever competition's numbers need reproducing.
- If picking up Section 10's top priority: an actual USB 80mm thermal receipt printer, its driver, and a Windows or Mac machine with a real display.

With those, a fresh session can:

- Re-run the pipeline against real CSVs (`calculations.pipeline.build_all(my_team_num=...)`) to confirm nothing has regressed.
- Confirm Stage 14/15/16 are present: five tabs ending in Match Simulator; a “Print Results” button next to “Go” on Match Lookup; a printing/ package with `escpos_builder.py` trimmed to the Stage-16 field set (Section 8); Data menu ending in “Select Receipt Printer...”.
- Cross-reference this PDF's Section 5 (stage history) and Sections 7–8 (printing detail) against the actual code before making further changes, rather than re-deriving the design from scratch.

End of summary. Generated to allow this project to be resumed with no access to the original chat history. This is the v10 revision, current through Stage 16 (Receipt Content Trim + Cut-off Fix).